INTERNATIONAL MONETARY FUND

LIBERIA

May 16, 2019

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

Text Table 1: Liberia: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited space to accommodate shocks*
Application of judgment	*No*

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Prepared by the International Monetary Fund and the World Bank.

The Debt Sustainability Analysis (DSA) suggests that Liberia remains at moderate risk of debt distress with limited space to accommodate shocks. The country's debt carrying capacity remains medium, but the rating has declined from 3.1 to 2.77. The authorities have pursued non-concessional loans, but none has been disbursed yet. The government has instead borrowed U.S. dollars from the Central Bank of Liberia (CBL) to close the financing gap in FY2018. Such new borrowing, as well as the legacy U.S. dollar debt from the civil war time, are both incorporated in the new DSA. The State-owned Enterprises (SOE) guaranteed debt is also incorporated. Liberia would edge closer to high risk of debt distress with a small change in the terms of both domestic and external debt or a failure to adjust primary expenditure to the available revenue envelope over the medium-term.

PUBLIC DEBT COVERAGE

1. The DSA covers central government debt, central government guaranteed debt, and central bank debt contracted on behalf of the government (Text Table 2).[1] The bulk of State-Owned Enterprise (SOE) debt is guaranteed by the central government and is included in DSA. Government borrowing from the Central Bank of Liberia (CBL) has been included in the current DSA analysis, consisting of $254.5 million (7.8 percent of GDP) in legacy U.S. debt from the war time and $65 million (1.5 percent of GDP) in credit from the CBL in the form of bridge loans and advances used to cover the fiscal financing gap of FY2018. In addition, the DSA includes $65 million in arrears to the construction sector, which took advances from banks to carry out public road projects. The largest debt of SOEs is a World Bank loan to the Liberia Electricity Corporation (LEC) for the rehabilitation of Mt. Coffee hydropower station. Other elements of public sector debt are not included in the analysis because of data constraints.[2]

Text Table 2: Liberia: Coverage of Public Sector Debt

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central government, central bank, government-guaranteed debt			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
4	PPP	35	percent of PPP stock	3.5	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			10.5	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

[1]The definition of external and domestic debt uses a residency criterion.

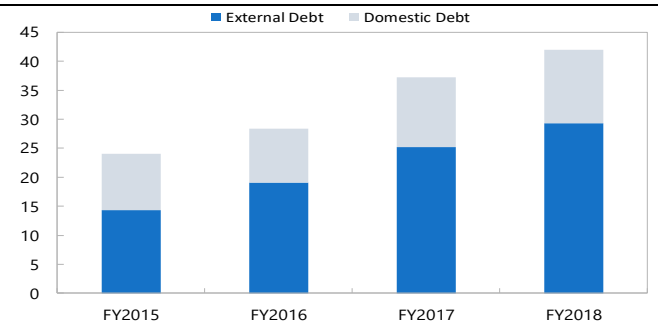
[2]The contingent liabilities shock from the SOE debt is kept at the default value of 2 percent to reflect risks associated with non-guaranteed SOE debt, currently excluded from the analysis due to the data availability constraints. Currently, the SOE Reporting and Coordination Unit (SOERCU) of the MFDP monitors and reports on the performance of 15 out of 39 registered SOEs in Liberia but the reports do not provide any specific information about non-guaranteed SOE debt. The amended PFM Act strengthens requirements for reporting and monitoring of SOE debt, including non-guaranteed. Going forward, the external debt coverage will be expanded as the government plans to include the SOE sector non-guaranteed debt into public sector debt.

BACKGROUND

2. This DSA is being conducted in the context of the 2019 Article IV consultation. The last Low-Income Country DSA (LIC-DSA) was considered by the Executive Board in June 2018 as part of the 2018 Article IV consultation.[3] Liberia does not currently have a Fund-supported program but continues to be subject to the IDA Non-Concessional Borrowing Policy (NCBP) regardless of the risk of debt distress.[4]

3. A year after taking office, policy uncertainty is besetting President George Weah's efforts to meet the expectations of the Liberian population. The government launched an ambitious pro-poor agenda aimed primarily at closing the country's infrastructure gap and raising the income level of the poor, but financing remains uncertain. On the macroeconomic front, large policy slippages have emerged, widening macroeconomic imbalances: i) the fiscal policy stance has been loose and coupled with big revenue shortfalls and financing gaps; ii) the monetary policy stance has been passively accommodative; and iii) the CBL has intervened in the foreign exchange market depleting international reserves. As a result, the medium-term outlook is less optimistic than at the time of the 2018 Article IV consultation.

Text Figure 1. Liberia: Stock of External Public and Publicly Guaranteed Debt, FY2015-18 [1/]
(Percent of GDP)



Sources: Liberian authorities and IMF staff calculations.
[1/]The debt to GDP ratios are calculated using external debt (in USD) evaluated at the end of period exchange rate over GDP (in USD) evaluated at the period average exchange rate.

[3]This DSA is prepared jointly by the staff of the IMF and World Bank, in collaboration with the authorities of Liberia. The current DSA follows the revised Debt Sustainability Framework (DSF) for LICs and Guidance Note (2017) in effect as of July 1, 2018. The last joint DSA was prepared for the 2018 Article IV consultation and can be found in IMF Country Report No. 18/172, June 2018.

[4]The NCBP requires a minimum grant element of 35 percent or higher, should a higher minimum be required under a Fund-supported program.

4. Identifying viable external financing to fulfill the pro-poor agenda has proved challenging. The authorities' efforts to secure financing from non-traditional donors, mostly on non-concessional terms, have not been successful. In a fast-tracked process that lacked transparency, the government of Liberia (GOL) ratified two separate external loan agreements totaling US$957.2 million (29 percent of GDP) in May of 2018.[5] More recently, the authorities have indicated that they are in the process of cancelling them. Yet, the authorities remain tolerant to the possibility of contracting large loans on non-concessional terms beyond the limits of Liberia's absorptive capacity. Meanwhile, external grants inflows are on a declining trend—expected to decrease from 16.7 percent of GDP in FY2017 to 13.4 percent of GDP in FY2024—and the authorities' capacity to mobilize domestic revenue remains limited for the size of the need. Therefore, a shrinking revenue envelope prevents the authorities' from effectively delivering the much-needed public services in a manner that is consistent with macroeconomic stability and conducive to long term growth.

Text Table 3. Liberia: Structure of External Public Debt as of June-2018[1/]

	USD millions	Percent of Total	Percent of GDP [2/]
Multilateral	**805**	**93.8**	**27.5**
IMF	231	26.9	7.9
World Bank	356	41.5	12.2
AfDB	92	10.7	3.1
EIB	55	6.4	1.9
Other Multilateral	72	8.4	2.5
Bilateral:Non-Paris Club	**54**	**6.2**	**1.8**
China	5	0.6	0.2
Kuwait	15	1.7	0.5
Saudi Arabia	34	3.9	1.1
Total	**859**	**100.0**	**29.3**

Sources: Liberian authorities and IMF staff calculations.

[1] Debt stock on disbursement base, excluding debt to Taiwan Province of China (85.4 mil) that is not servicing due to the lack of diplomatic relationship.

[2] The debt to GDP ratios are calculated using external debt (in USD) evaluated at the end of period exchange rate over GDP (in USD) evaluated at the period average exchange rate, to ensure consistency with the DSA template.

5. Following the completion of HIPC in 2010, debt accumulated rapidly due to scaled-up infrastructure spending and the government's response to a series of adverse shocks. The total public external debt stock was $859 million (29.3 percent of GDP)[6] at end-FY2018, comprising mostly of multilateral loans (Text Table 3). The recent uptick in domestic debt in the form of credit from the CBL raises concerns about macroeconomic stability as the Central Bank is absorbing the losses of the lending extended to the GOL. Total public and publicly guaranteed debt at end-FY2018 is estimated at 42 percent of GDP (Text Figure 1).

[5]The agreements were signed between the Government of Liberia and Eton Finance PTE limited for US$536 million and the EBOMAF company of Burkina Faso for US$420.81 million. Projects targeted under the financing agreements included the construction of 770 kilometers of asphalt roads, seven local football stadiums, and a vocational training center.

[6]The ratio is calculated using external debt (in USD) evaluated at the end of period exchange rate over GDP (in USD) evaluated at the period average exchange rate.

UNDERLYING ASSUMPTIONS

6. The key macroeconomic assumptions are in line with the baseline scenario described in the Policy Note. The near- and medium-term outlook is challenging and less optimistic than at the time of the 2018 Article IV consultation. Changes to the underlying assumptions are as following (Text Table 4):

- The ***real GDP growth path*** has been revised down from 4.7 percent to 0.4 percent in 2019 due to continued policy uncertainty and slowing down of credit growth. Under the baseline scenario the authorities are expected to muddle through initially but face a forced adjustment in year 2022. Growth in the short-to-medium term is supported by agriculture and mining. Real GDP is projected to contract by 1.3 percent in 2022 and 0.5 percent in 2023. Medium-term growth has been revised down from 5.3 percent to 3.7 percent in line with factor accumulation (Annex I in Staff Report). In the long-run, growth is estimated to increase to 4.3 percent by 2028—owing mostly to improvements in total factor productivity and growth of the working-age population—and remaining stable thereafter.

Text Table 4. Liberia: Underlying DSA Assumptions

	FY2019	FY2019-24	FY2024-40
Nominal GDP	(In million US dollar)	Average growth (In percent)	Average growth (In percent)
June 2018 Board	3442	6.8	7.8
Current	3267	1.1	6.9
Exports of Goods and Service			
June 2018 Board	727	5.0	6.1
Current	777	4.2	5.8

Sources: Liberian authorities; and IMF staff projections.

- The ***inflation*** outlook deteriorated significantly in the second half of 2018. Closely following developments in the foreign exchange markets, average headline inflation increased to 20.4 percent in 2018 (compared to 11.7 estimated previously) and is expected to remain in double digits in the medium-term.

- The ***fiscal position*** of the central government is set to remain loose. The fiscal deficit widened to 5.5 percent of GDP in FY2018, partially financed by credit from the central bank. An overoptimistic revenue envelope driven by the pressure to address high spending needs means that—in the absence of additional budget support—the authorities will continue to rely on borrowing from the central bank in both U.S. and Liberian dollars to close the financing gap.

- The ***current account deficit*** estimates for 2019 widened from 22.3 to 23.3 percent of GDP. The trade balance improved modestly in 2018, but it was more than offset by a decline in net transfers following the United Nations Mission in Liberia (UNMIL) withdrawal and a decline in net income. Over the medium term, the current account deficit is expected to remain high during the next few years but then decline in subsequent years in response to the forced adjustment. The external sector assessment shows that Liberia's external sector position is substantially weaker than implied by fundamentals and desirable policies.

- The CBL's foreign ***reserves*** continued to decline. Staff estimate that the CBL will use around US$80 million of its reserves to finance fiscal deficits, its operations, and interventions in the foreign exchange market in 2019. They are expected to decline in the medium-term to below 1 months of imports.

7. The assumptions for the financing mix and borrowing terms are as follows:

- ***External borrowing***. The DSA assumes that public external debt would increase by about $1.24 billion in the medium-term. Even if the two external loans ratified in May 2018 (for US$957.5 million) were to be officially canceled, the baseline assumes large non-concessional loans. The average grant element of new borrowing is estimated to decline from 43.6 percent FY2019 to 31.2 percent in FY2022, and subsequently increasing as the authorities increase engagement with and borrow more from traditional donors. This is in contrast with an average grant element of the new borrowing of 45 percent assumed at the time of the 2018 Article IV consultation.

- ***Domestic borrowing***. The baseline scenario assumes central bank credit during 2019–21 at a negative real interest rate to fulfil central government's budget needs (e.g., the government currently pays -2.2 percent in real terms on the legacy debt of $250 million and zero nominal interest rate on the credit extended in 2018). The baseline also assumes that the financing gap will be filled with additional central bank financing. As the central government faces high spending pressure, amortization of CBL credit for budget support will be deferred to the long-term.

REALISM OF THE BASELINE ASSUMPTIONS

8. Drivers of the debt dynamics seem to reflect the past well (Figure 3).

- The DSF shows the evolution of the past and projections of external and public debt to GDP ratios for the current DSA, the previous year DSA (the 2018 Article IV DSA), and the DSA from 5 years past. The current DSA shows a significantly different evolution from the past, which reflects the latest revisions to the medium-term outlook and policy direction of the authorities. The downward revisions to real GDP growth explain most of the increase in the ratio of external-debt-to GDP in coming years. Some additional financing on non-concessional terms was added relative to the 2018 Article IV DSA, as the authorities have continued to seek financing from non-traditional donors, mostly on non-concessional terms, though non-financing has been disbursed yet. The inclusion of CBL credit in the current DSA and the assumption that the financing gap will be filled with additional CBL credit explain the differences in domestic debt dynamics outlined in the previous DSA.

- A high contribution of unexpected non-interest current account deficits to past debt accumulation and an equally large unexpected residual to the past debt accumulation in the opposite direction are observed (Figure 3). These debt dynamics are plausible since residual financing (i.e., net private financing under other investment flows in the Balance of

Payments, Table 2), which is enabling the large current account deficit, includes current transfers (remittances) that are not captured by the official statistics. In other words, the residual financing shown in the realism tool is unlikely to have been debt creating flows. Indeed, the Other Depository Corporations (ODCs) survey does not show such an accumulation.

- The significant adverse shocks the country has gone through (the Ebola epidemic and the commodity price shock) are well captured in the chart showing unexpected changes in debt accumulation in the past 5 years. The unexpected increase in debt was about 12 percent of GDP, which is above the median of the countries producing LIC DSA and is towards the upper end of the interquartile range (25 percent–75 percent). As expected in countries that go through major macroeconomic shocks, drivers of this unexpected debt accumulation are equally shared among three factors: unexpected increases in primary fiscal deficits, an unexpected decline in growth, and an unexpected depreciation of the real exchange rate.

9. The widening of the primary fiscal deficit in the next three years is within the range of countries producing the LIC DSA.

- The second DSF realism tool assesses the realism of the fiscal projection. The fiscal primary deficit is projected to increase by about 1.0 percentage point of GDP in the next three years. This is just below the mode.

10. The consistency of the fiscal deficit and growth projections for 2019 and 2020 is evident in Figure 6 for a couple of reasons.

- The growth path shown in the figure captures overall growth, but the impact of fiscal adjustment/slippage is assumed to impact the non-mining sector growth. The fiscal multiplier used is 0.67.
- The fiscal widening in 2019 is projected to have a small stimulus impact relative to the adverse impact of the terms-of-trade shocks and the sharp contraction of imports of key commodities (fuel and rice) (Figure 3).

11. The growth and fiscal adjustment tool (Figure 5) shows that growth projections for the 2018 Article IV DSA were on the optimistic side, whereas the downward revision to growth reflected in the current DSA is in line with past experience.

COUNTRY CLASSIFICATION AND MODEL SIGNAL

12. Liberia's debt carrying capacity is classified as medium, based on a Composite Indicator (CI) of 2.77 (Text Table 4).[7] The country's capacity was assessed at the same level in the

[7]The CI captures the impact of the different factors through a weighted average of the World Bank's 2017 Country Policy and Institutional Assessment (CPIA) score, the country's real GDP growth, remittances, international reserves, and world growth. A country's debt-carrying capacity would be assessed as weak if its CI value is below 2.69, medium if it lies between 2.69 and 3.05, and strong if it is above 3.05.

previous DSA (Text Table 5). However, subsequent revisions to the deteriorating macroeconomic outlook could affect the CI in the next 1–2 years, significantly reducing the borrowing space. In addition, Liberia was recently downgraded to "weak quality of debt monitoring" in line with the country's debt recording capacity.

13. Standard scenarios stress test and a contingent liability test are conducted and discussed below.

Text Table 4. Liberia: Composite Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.103	1.19	43%
Real growth rate (in percent)	2.719	3.358	0.09	3%
Import coverage of reserves (in percent)	4.052	21.268	0.86	31%
Import coverage of reserves^2 (in percent)	-3.990	4.523	-0.18	-7%
Remittances (in percent)	2.022	15.020	0.30	11%
World economic growth (in percent)	13.520	3.660	0.49	18%
CI Score			2.77	100%
CI rating			Medium	

Text Table 5. Liberia: Debt Carrying Capacity and Thresholds

Final Classification	Classification based on current vintage	Classification based on the previous LIC DSF
Medium	Medium	Medium
	2.8	3.1

EXTERNAL debt burden thresholds	Current LIC DSF	Previous LIC DSF
PV of debt in % of		
Exports	180	100
GDP	40	30
Debt service in % of		
Exports	15	15
Revenue	18	18
TOTAL public debt benchmark	**Current LIC DSF**	**Previous LIC DSF**
PV of total public debt in percent of GDP	55	38

EXTERNAL DSA

14. Liberia remains at moderate risk of external debt distress. The PV of debt-to-GDP and the PV of debt-to-export ratios are set to remain below their policy dependent threshold, with the former peaking at 34.6 percent in 2027 and the later remaining below 150 percent in the medium-to long-term (Figure 1). The debt-service to revenue ratio will reach levels just below the threshold of 18 percent in 2023 and 2024 potentially crowding out much needed spending and running the risk

of a buildup of arrears. The debt-service to exports ratio remains below the threshold, peaking in 2023 and 2024 when the economy is forced to adjust. The evolution of the probability of debt distress over time, shows that the probability of debt distress increases substantially if there is a shock to exports or depreciation (Figure 6).

15. Standard stress tests show that a further deterioration of the macroeconomic outlook will lead to breaches of the policy dependent thresholds (Table 3). All standard stress tests, namely, a shock of one-standard deviation in the primary balance, nominal export growth, other non-debt creating flows, and a one-time depreciation of the size needed to close the real exchange overvaluation (which is the most extreme shock) will all result in breaching the thresholds on the stock of debt (the PV of debt-to-GDP ratio and the PV of debt-to-exports ratio). A one-standard deviation shock to real GDP growth will not breach the threshold on the stock of debt, but it will breach the threshold on debt services.

PUBLIC DSA

16. Public sector debt indicators also show that Liberia remains at moderate risk of public debt distress. The PV of public debt-to GDP ratio is projected to increase from an estimate of 35.3 percent in FY2019 to 46.6 percent in FY2023 and continue on an upward trend thereafter (Table 2 and Figure 2). The PV of debt-to-revenue ratio will peak at 168.5 percent in FY2025 and decline slowly to 156.9 percent in the long-term, while the debt-service-to-revenue ratio will peak at 16.8 percent in FY2023 and decline in subsequent years.

17. Under standard sensitivity analysis, the PV of Debt-to-GDP breaches the relevant threshold. Based on standard stress tests, a deterioration of real GDP, the primary balance, exports, and other flows results in a breach of the threshold on the PV of debt-to-GDP ratio (Table 4). Moreover, the debt service-to-revenue ratio reaches close to 20 percent (or above 20 percent in some cases). Additionally, the contingent liability stress test is estimated to lead to a one-off increase in the debt-to-GDP ratio of 10.5 percent, capturing the combined shock of SOE's external debt default, PPPs' distress, and financial market vulnerabilities that are not included in the covered data.

18. An alternative scenario in which an increase in interest payments does not crowd out primary expenditure is also considered. A case in which debt service increases because of a rise in the interest rate or additional borrowing that must be serviced, but the GOL does not adjust primary expenditure accordingly is analyzed. Under this scenario, Liberia will eventually breach the PV of debt-to-GDP ratio if additional revenue is not mobilized and the GOL is forced to borrow to meet its increased debt service obligations.

RISK RATING AND VULNERABILITIES

19. Liberia's borrowing space is limited, and careful consideration to the terms of new external borrowing and the country's absorptive capacity is needed (Figure 4). The authorities' ambitious infrastructure program to rehabilitate the national road network will undoubtedly raise

the PV of debt relative to its foreign exchange earning capacity. The projected disbursement path for the period of FY2019–24 is beyond what the authorities have been able to absorb in the past but reflects their ambition to secure large infrastructure loans that promise to be disbursed in the short-term. Yet, under this scenario, the limited space to absorb shocks is almost non-existent for debt-service-to-revenue ratio. Moreover, borrowing beyond capacity will not generate enough GDP growth to compensate for the increase in nominal debt levels. Indeed, the effect of the rehabilitation of roads on aggregate demand from the financing options under consideration is expected to be limited, as only a small part of the total cost of rehabilitation of the roads would be likely to be sourced locally.

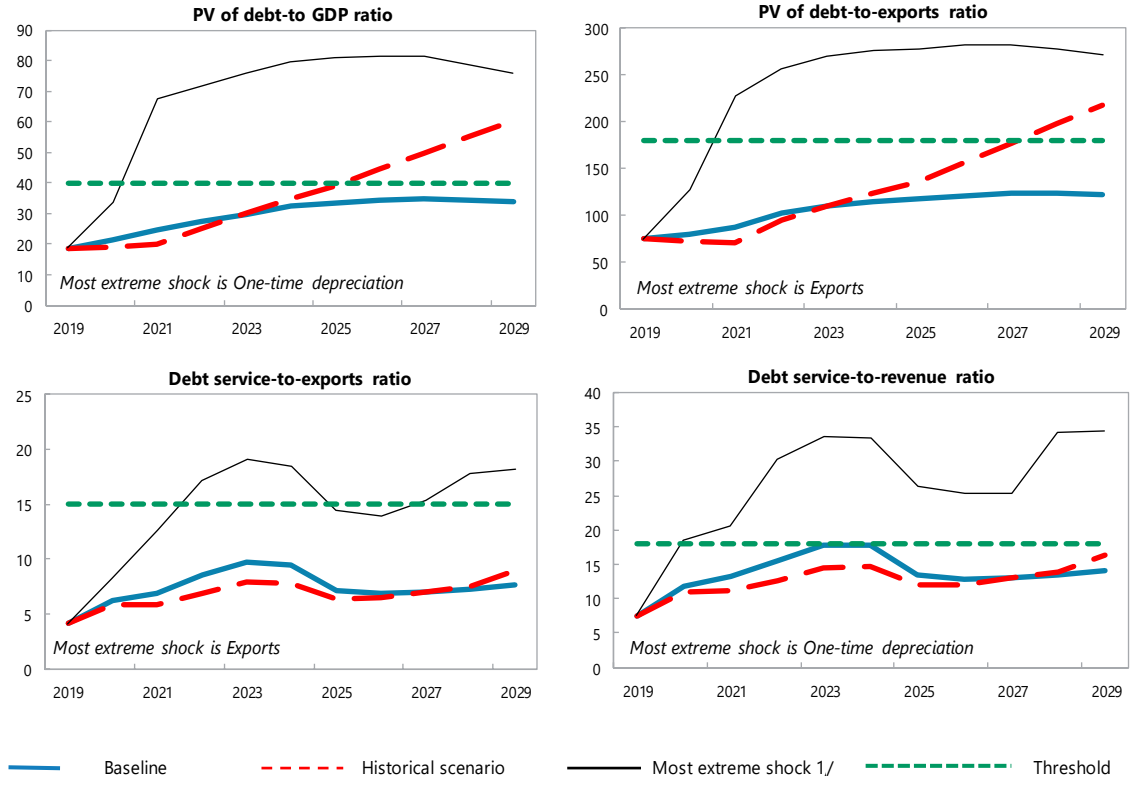
20. Moreover, while most of the domestic debt is intra-government borrowing, fiscal adjustment is warranted to ensure macroeconomic stability and solvency of the Central Bank. Currently, the CBL is absorbing the losses of the credit it is providing to the GOL and no repayment plan has been agreed, while the resources of the Central Bank are being depleted. A repayment profile of domestic debt with a positive real interest rate will put additional pressure on spending needs, crowding out primary expenditure or leading to a further increase of the fiscal imbalance.

21. Risks to the outlook are titled to the downside. On the upside, an increase in commodity prices, an increase in iron ore production, and an increase in donor grants could ease some of the macroeconomic pressures. Moreover, if the non-concessional borrowing planned in the baseline scenario beyond absorptive capacity does not materialize debt will remain at more sustainable levels. However, on the downside, a drop-in commodity prices, or a failure to mobilize fiscal resources to close the financing gap could lead to a further deterioration of macroeconomic conditions and increase the risk of debt distress.

AUTHORITIES' VIEWS

22. The authorities broadly agreed with the importance of maintaining debt sustainability in the medium term. The authorities reiterated their preference for concessional financing as it recognizes that borrowing space is limited and sensitive to the terms of new loans; and noted that they are in the process of cancelling the non-concessional ETON and EBOMAF loans. However, given its stated commitment to fulfill its Pro-Poor Agenda, if assistance from traditional donors is delayed, it remains tolerant to the possibility of contracting non-concessional loans and securing financing from non-traditional donors. The authorities emphasized the pressing need to move forward with their development agenda, and expressed the hope that the international community would provide the current administration with assistance in meeting these needs through provision of budget support, project grants, and financing for infrastructure projects.

Figure 1. Liberia: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2019–29



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.1%	2.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	6	6

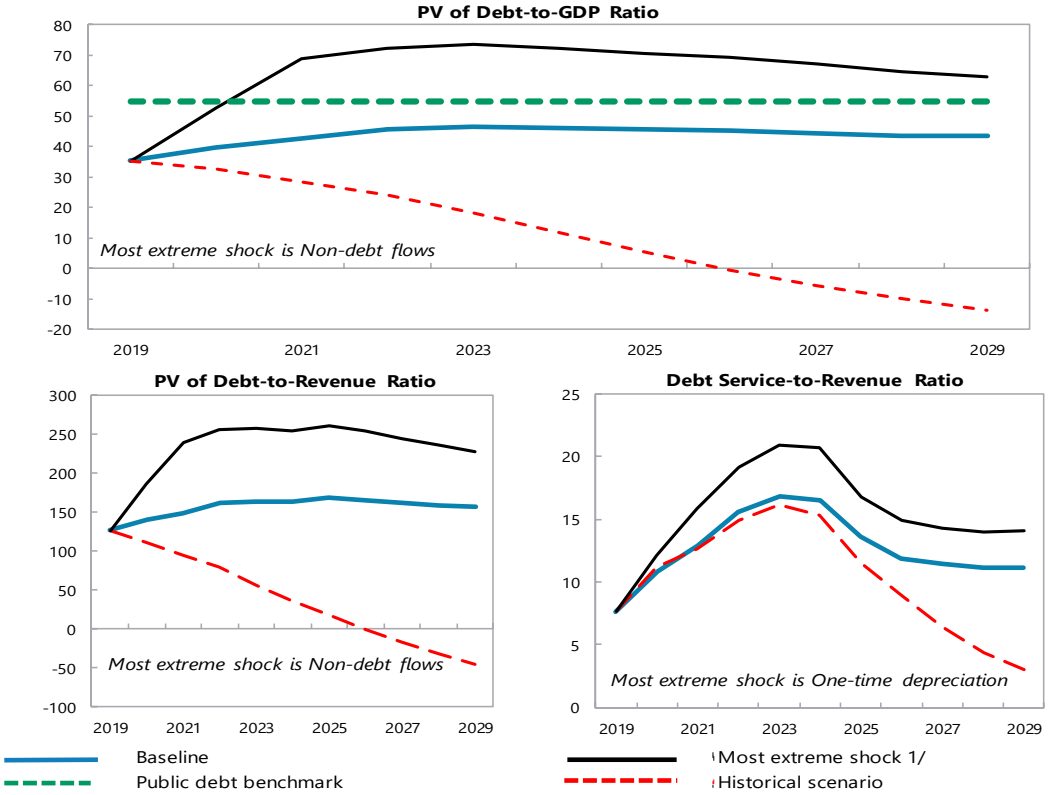
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Liberia: Indicators of Public Debt Under Alternative Scenarios, 2019–29



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	82%	82%
Domestic medium and long-term	18%	18%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.1%	2.1%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	-2.2%	-2.2%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	0.0%	0.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

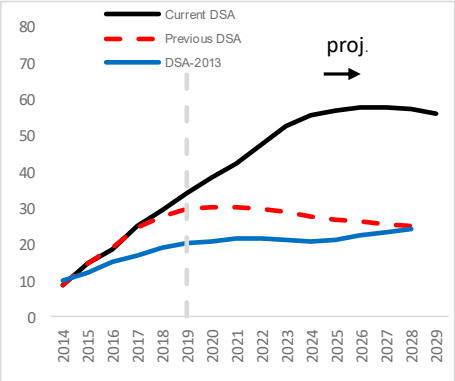
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

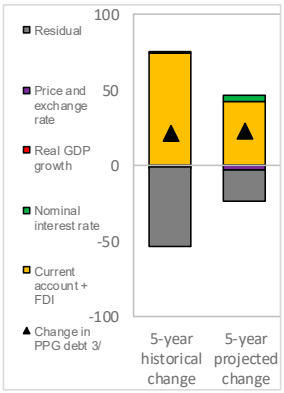
Figure 3. Liberia: Drivers of Debt Dynamics – Baseline Scenario

External debt

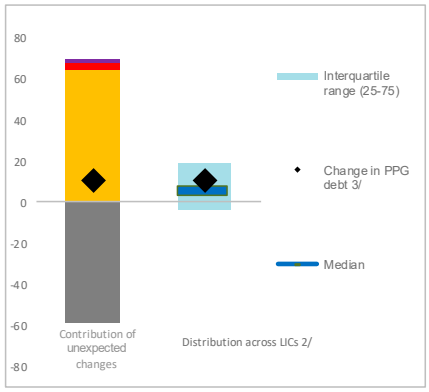
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

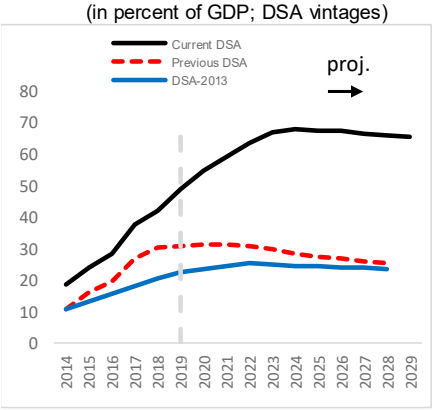


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

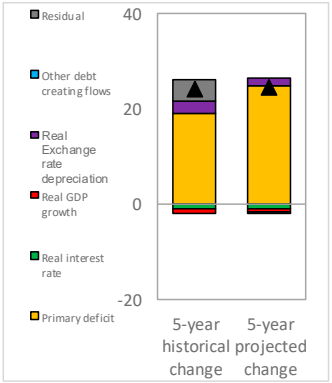


Public debt

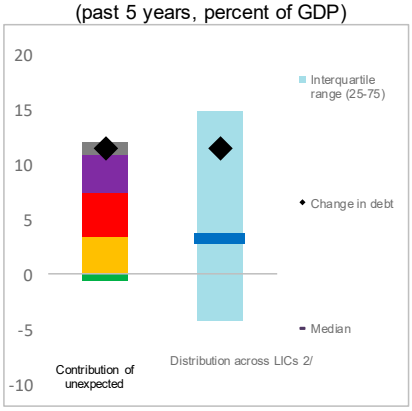
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

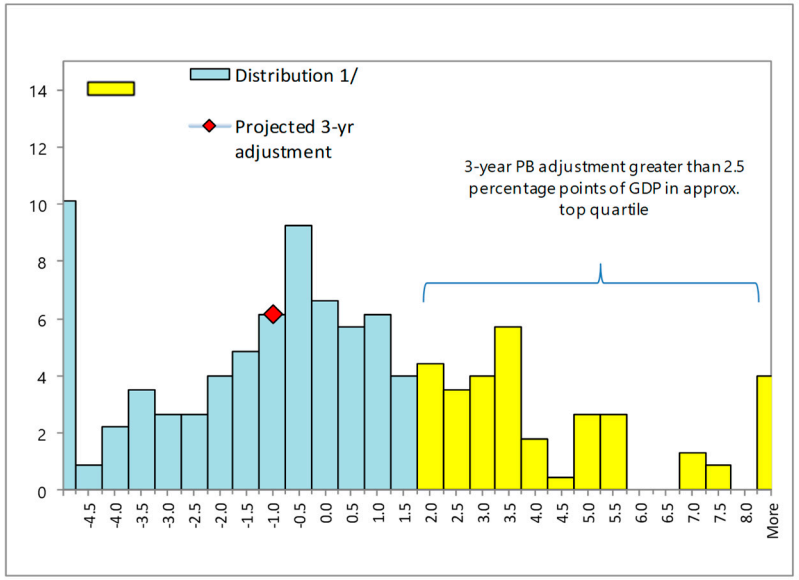
Figure 4. Liberia: Qualification of the Moderate Category, 2019–29[1/]

Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

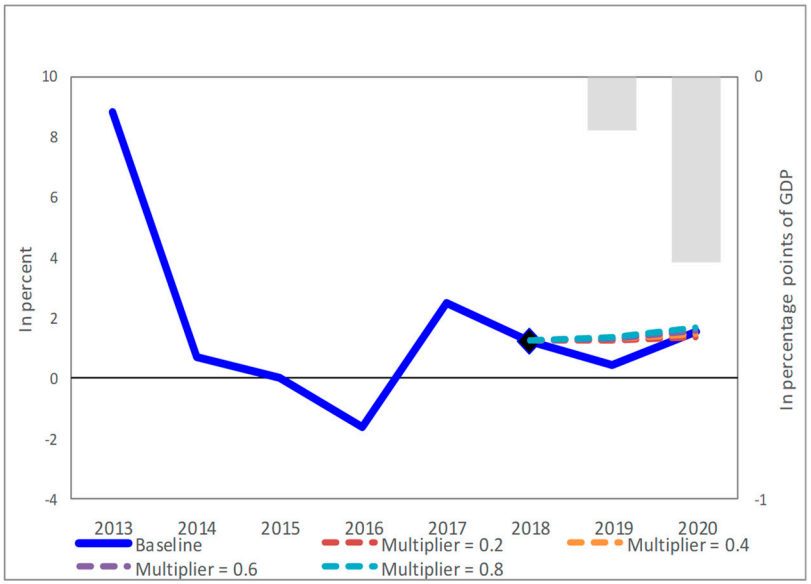
Figure 5. Liberia: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



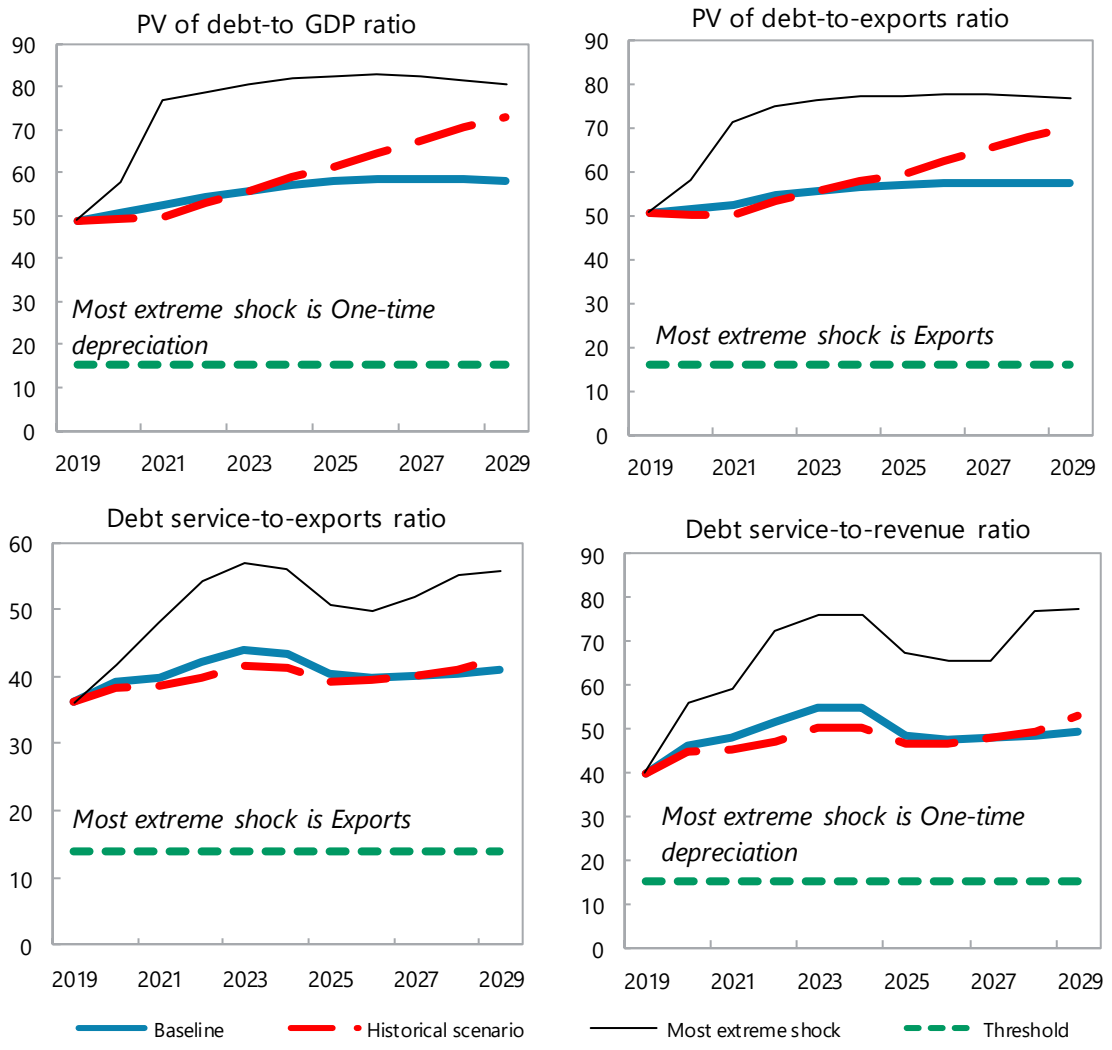
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths[1/]



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Figure 6. Liberia: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 1/



Sources: Country authorities; and staff estimates and projections.

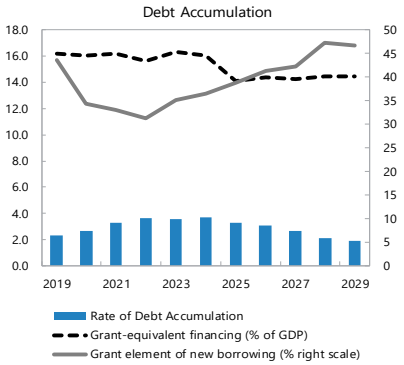
1/ The probability approach focuses on the evolution of the probability of debt distress over time, rather than on the evolution of debt burden indicators.

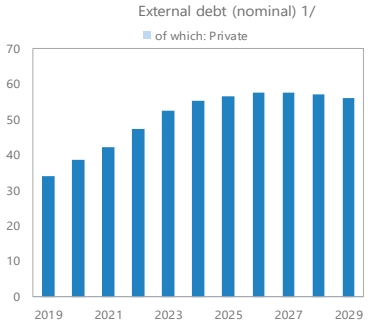
Table 1. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2016–39

(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**18.8**	**25.1**	**29.3**	**34.1**	**38.6**	**42.2**	**47.5**	**52.5**	**55.4**	**56.0**	**39.2**	**12.4**	**50.5**
of which: public and publicly guaranteed (PPG)	**18.8**	**25.1**	**29.3**	**34.1**	**38.6**	**42.2**	**47.5**	**52.5**	**55.4**	**56.0**	**39.2**	**12.4**	**50.5**
Change in external debt	4.1	6.3	4.1	4.8	4.5	3.6	5.3	5.0	2.9	-1.1	-1.5		
Identified net debt-creating flows	**11.0**	**16.1**	**14.4**	**12.5**	**10.6**	**11.0**	**5.6**	**6.4**	**3.6**	**-1.9**	**-4.4**	**9.7**	**4.6**
Non-interest current account deficit	**18.7**	**23.5**	**23.0**	**22.7**	**22.3**	**23.9**	**15.8**	**15.8**	**15.6**	**10.2**	**5.6**	**21.6**	**16.3**
Deficit in balance of goods and services	-97.4	-82.4	-76.1	-75.2	-77.5	-80.7	-71.3	-72.6	-75.1	-71.3	-60.8	**-112.8**	**-74.6**
Exports	23.9	22.7	23.8	25.1	26.9	28.0	26.8	27.1	28.0	27.6	25.0		
Imports	-73.5	-59.7	-52.3	-50.1	-50.7	-52.6	-44.5	-45.5	-47.1	-43.7	-35.8		
Net current transfers (negative = inflow)	-37.2	-22.7	-18.8	-18.3	-18.5	-18.5	-18.4	-18.4	-18.1	-15.4	-10.5	**-39.9**	**-17.5**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	153.4	128.6	118.0	116.2	118.4	123.1	105.5	106.8	108.8	96.9	76.9	**174.3**	**108.4**
Net FDI (negative = inflow)	**-7.2**	**-7.5**	**-8.8**	**-10.6**	**-12.0**	**-13.1**	**-11.7**	**-11.1**	**-11.7**	**-10.9**	**-8.9**	**-12.5**	**-11.6**
Endogenous debt dynamics 2/	**-0.4**	**0.1**	**0.1**	**0.3**	**0.3**	**0.2**	**1.5**	**1.7**	**-0.3**	**-1.2**	**-1.1**		
Contribution from nominal interest rate	0.1	0.2	0.3	0.5	0.8	0.6	1.0	1.5	1.6	1.1	0.5		
Contribution from real GDP growth	0.2	-0.5	-0.3	-0.1	-0.5	-0.5	0.5	0.2	-1.9	-2.3	-1.7		
Contribution from price and exchange rate changes	-0.8	0.4	0.1	...	...	...	...	...	...	...	...		
Residual 3/	**-6.9**	**-9.8**	**-10.2**	**-7.7**	**-6.1**	**-7.4**	**-0.3**	**-1.4**	**-0.7**	**0.8**	**2.9**	**-6.8**	**-2.2**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**16.2**	**18.7**	**21.5**	**24.6**	**27.3**	**29.9**	**32.3**	**33.7**	**24.7**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**68.3**	**74.7**	**80.0**	**87.8**	**102.1**	**110.3**	**115.2**	**122.1**	**98.6**		
PPG debt service-to-exports ratio	**0.7**	**1.3**	**2.8**	**4.1**	**6.3**	**6.8**	**8.5**	**9.7**	**9.4**	**7.6**	**9.3**		
PPG debt service-to-revenue ratio	**1.2**	**2.1**	**5.1**	**7.5**	**11.9**	**13.2**	**15.6**	**17.7**	**17.7**	**14.0**	**15.4**		
Gross external financing need (Million of U.S. dollars)	375.8	530.8	487.5	427.4	386.6	414.0	213.2	255.1	235.2	73.4	-92.9		
Key macroeconomic assumptions													
Real GDP growth (in percent)	-1.6	2.5	1.2	0.4	1.6	1.3	-1.3	-0.5	3.7	4.3	4.3	**3.9**	**2.4**
GDP deflator in US dollar terms (change in percent)	5.8	-1.9	-0.5	-1.4	-2.0	-0.5	4.7	4.0	0.2	2.8	2.1	**3.8**	**1.6**
Effective interest rate (percent) 4/	0.9	0.9	1.2	1.6	2.3	1.7	2.5	3.2	3.2	2.0	1.4	**1.1**	**2.4**
Growth of exports of G&S (US dollar terms, in percent)	-5.4	-4.6	5.7	4.3	6.6	5.2	-1.3	4.7	7.6	6.2	4.3	**4.5**	**5.4**
Growth of imports of G&S (US dollar terms, in percent)	-15.2	-18.3	-11.8	-5.1	0.6	4.6	-12.5	5.8	7.6	5.2	5.0	**-0.5**	**2.5**
Grant element of new public sector borrowing (in percent)	...	...	...	43.6	34.3	33.0	31.2	35.1	36.5	46.7	32.7	**...**	**39.1**
Government revenues (excluding grants, in percent of GDP)	14.0	14.3	13.0	13.7	14.1	14.5	14.7	14.8	14.9	15.1	15.1	**14.7**	**14.8**
Aid flows (in Million of US dollars) 5/	736.6	663.4	549.7	556.4	566.4	584.2	589.4	653.0	671.0	772.5	1416.1		
of which: Grants	623.6	541.6	425.5	461.3	457.9	460.1	453.9	478.6	482.7	627.8	1266.4		
of which: Concessional loans	113.0	121.8	124.2	95.1	108.6	124.1	135.5	174.5	188.3	144.8	149.7		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	16.2	16.0	16.2	15.6	16.3	16.1	14.4	13.7	**...**	**15.3**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	86.5	82.4	79.9	77.5	77.8	77.6	87.0	87.8	**...**	**82.0**
Nominal GDP (Million of US dollars)	3,226	3,244	3,267	3,235	3,219	3,241	3,352	3,468	3,606	5,010	9,913		
Nominal dollar GDP growth	4.0	0.6	0.7	-1.0	-0.5	0.7	3.4	3.5	4.0	7.2	6.6	**7.9**	**4.0**
Memorandum items:													
PV of external debt 7/	...	...	16.2	18.7	21.5	24.6	27.3	29.9	32.3	33.7	24.7		
In percent of exports	...	...	68.3	74.7	80.0	87.8	102.1	110.3	115.2	122.1	98.6		
Total external debt service-to-exports ratio	0.7	1.3	2.8	4.1	6.3	6.8	8.5	9.7	9.4	7.6	9.3		
PV of PPG external debt (in Million of US dollars)			530.4	605.3	691.7	797.9	915.4	1035.5	1164.1	1690.3	2446.6		
(PVt-PVt-1)/GDPt-1 (in percent)				2.3	2.7	3.3	3.6	3.6	3.7	1.9	0.8		
Non-interest current account deficit that stabilizes debt ratio	14.6	17.2	18.9	17.9	17.8	20.3	10.4	10.8	12.7	11.3	7.1		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

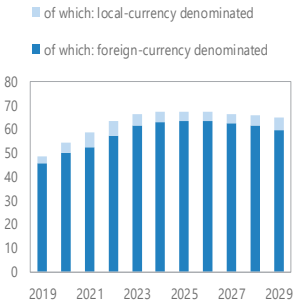
Table 2. Liberia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016–39

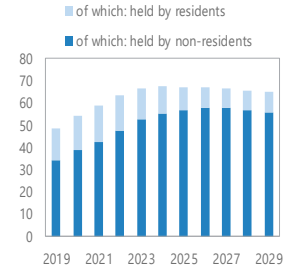
(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**28.1**	**37.4**	**42.0**	**48.5**	**54.4**	**58.7**	**63.3**	**66.6**	**67.6**	**65.1**	**75.3**	**24.2**	**62.8**
of which: external debt	**18.8**	**25.1**	**29.3**	**34.1**	**38.6**	**42.2**	**47.5**	**52.5**	**55.4**	**56.0**	**39.2**	**12.4**	**50.5**
Change in public sector debt	4.1	9.3	4.6	6.5	5.9	4.3	4.6	3.3	1.0	-0.6	2.3		
Identified debt-creating flows	**2.9**	**5.7**	**5.3**	**6.3**	**6.0**	**4.3**	**4.7**	**3.5**	**1.2**	**-0.8**	**2.3**	**-5.2**	**2.1**
Primary deficit	**2.4**	**4.5**	**4.9**	**5.1**	**5.5**	**5.8**	**4.4**	**4.1**	**4.0**	**2.6**	**5.7**	**-6.4**	**3.8**
Revenue and grants	33.3	31.0	26.0	28.0	28.4	28.7	28.2	28.6	28.3	27.6	27.9	**23.2**	**27.9**
of which: grants	19.3	16.7	13.0	14.3	14.2	14.2	13.5	13.8	13.4	12.5	12.8		
Primary (noninterest) expenditure	35.7	35.5	30.9	33.0	33.9	34.5	32.7	32.7	32.2	30.2	33.6	**16.8**	**31.7**
Automatic debt dynamics	**0.5**	**1.2**	**0.4**	**1.3**	**0.5**	**-1.5**	**0.3**	**-0.6**	**-2.8**	**-3.4**	**-3.4**		
Contribution from interest rate/growth differential	0.3	-0.8	-1.0	-0.4	-0.7	-1.1	0.3	0.3	-1.9	-2.7	-3.4		
of which: contribution from average real interest rate	-0.1	-0.2	-0.6	-0.2	0.0	-0.5	-0.4	0.0	0.5	0.0	-0.3		
of which: contribution from real GDP growth	0.4	-0.7	-0.5	-0.2	-0.7	-0.7	0.8	0.3	-2.4	-2.7	-3.0		
Contribution from real exchange rate depreciation	0.3	2.1	1.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**1.1**	**3.6**	**-0.7**	**1.8**	**1.2**	**-0.4**	**-0.2**	**-1.2**	**-1.1**	**-0.5**	**0.0**	**7.8**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**30.8**	**35.3**	**39.6**	**42.8**	**45.6**	**46.6**	**46.2**	**43.4**	**61.2**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**118.5**	**126.3**	**139.5**	**149.3**	**161.6**	**163.1**	**163.3**	**156.9**	**219.4**		
Debt service-to-revenue and grants ratio 3/	**1.6**	**1.4**	**3.8**	**7.7**	**10.8**	**12.9**	**15.5**	**16.8**	**16.5**	**11.2**	**18.8**		
Gross financing need 4/	2.9	4.9	5.9	7.2	8.6	9.5	8.8	8.9	8.6	5.7	10.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-1.6	2.5	1.2	0.4	1.6	1.3	-1.3	-0.5	3.7	4.3	4.3	**3.9**	**2.4**
Average nominal interest rate on external debt (in percent)	0.9	0.9	1.3	1.7	2.6	1.9	2.7	3.5	3.4	2.0	1.4	**1.1**	**2.5**
Average real interest rate on domestic debt (in percent)	-9.0	-9.6	-15.9	-15.4	-11.3	-6.7	-9.5	-10.9	-3.9	0.0	-0.3	**-9.1**	**-6.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	1.1	7.6	3.9	...	...	...	...	...	...	...	...	**-0.8**	**...**
Inflation rate (GDP deflator, in percent)	11.7	12.6	21.9	24.5	22.5	17.9	23.0	24.8	15.6	7.0	5.3	**11.4**	**16.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	-3.0	1.8	-11.8	7.3	4.0	3.2	-6.6	-0.3	2.2	5.5	5.4	**2.3**	**2.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.7	-4.8	0.4	-1.4	-0.4	1.5	-0.2	0.8	3.0	3.2	3.4	**-2.0**	**1.7**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019–29

(Percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	19	21	25	27	30	32	33	34	35	34	34
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	19	19	20	25	30	35	39	**44**	**50**	**55**	**60**
B. Bound Tests											
B1. Real GDP growth	19	23	28	31	34	37	38	39	39	39	38
B2. Primary balance	19	28	37	**40**	**43**	**46**	**47**	**48**	**47**	**47**	**46**
B3. Exports	19	27	40	**43**	**46**	**48**	**49**	**50**	**50**	**48**	**47**
B4. Other flows 3/	19	33	**49**	**52**	**54**	**57**	**58**	**58**	**57**	**55**	**53**
B5. Depreciation	19	33	**68**	**72**	**76**	**80**	**81**	**82**	**81**	**79**	**76**
B6. Combination of B1-B5	19	32	**49**	**52**	**55**	**57**	**58**	**59**	**58**	**56**	**54**
C. Tailored Tests											
C1. Combined contingent liabilities	19	27	31	34	36	39	40	**41**	**41**	**40**	40
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	75	80	88	102	110	115	118	121	123	123	122
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	75	72	70	94	111	124	137	157	177	**198**	**218**
B. Bound Tests											
B1. Real GDP growth	75	80	88	102	110	115	118	121	123	123	122
B2. Primary balance	75	103	132	150	160	164	165	168	169	167	165
B3. Exports	75	128	**228**	**257**	**271**	**276**	**278**	**283**	**283**	**278**	**272**
B4. Other flows 3/	75	125	175	**194**	**201**	**203**	**203**	**205**	**203**	**197**	**191**
B5. Depreciation	75	80	155	173	**180**	**183**	**183**	**185**	**186**	**182**	177
B6. Combination of B1-B5	75	131	161	**235**	**245**	**248**	**248**	**251**	**249**	**243**	**236**
C. Tailored Tests											
C1. Combined contingent liabilities	75	102	110	126	134	139	141	144	145	145	144
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	4	6	7	9	10	9	7	7	7	7	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	4	6	6	7	8	8	6	6	7	8	9
B. Bound Tests											
B1. Real GDP growth	4	6	7	9	10	9	7	7	7	7	8
B2. Primary balance	4	6	7	10	11	11	8	8	9	10	11
B3. Exports	4	8	13	17	19	18	14	14	15	18	18
B4. Other flows 3/	4	6	8	11	12	12	10	9	11	13	13
B5. Depreciation	4	6	7	11	12	11	9	9	9	12	12
B6. Combination of B1-B5	4	7	10	14	16	15	12	12	14	16	16
C. Tailored Tests											
C1. Combined contingent liabilities	4	6	7	9	10	10	8	7	8	8	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	8	12	13	16	18	18	13	13	13	13	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	8	11	11	13	15	15	12	12	13	14	16
B. Bound Tests											
B1. Real GDP growth	8	13	15	18	**20**	**20**	15	15	15	15	16
B2. Primary balance	8	12	15	**18**	**20**	**20**	16	15	17	**19**	**20**
B3. Exports	8	12	15	**19**	**22**	**22**	17	16	18	**20**	**21**
B4. Other flows 3/	8	12	16	**21**	**23**	**22**	18	17	**21**	**24**	**24**
B5. Depreciation	8	**19**	**21**	**30**	**33**	**33**	**26**	**25**	**25**	**34**	**34**
B6. Combination of B1-B5	8	12	16	**21**	**24**	**23**	**19**	18	**21**	**25**	**25**
C. Tailored Tests											
C1. Combined contingent liabilities	8	12	14	17	**19**	**19**	15	14	14	14	15
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Liberia: Sensitivity Analysis for Key Indicators of Public Debt, 2019–29

(Percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	35	40	43	46	47	46	46	45	44	42	41
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	35	33	28	24	18	12	5	0	-6	-11	-15
B. Bound Tests											
B1. Real GDP growth	35	41	48	51	53	54	55	**55**	**55**	55	55
B2. Primary balance	35	45	**56**	**59**	**59**	**59**	**58**	**57**	**55**	54	52
B3. Exports	35	44	**56**	**59**	**60**	**60**	**59**	**57**	**55**	53	51
B4. Other flows 3/	35	53	**69**	**72**	**74**	**72**	**71**	**69**	**67**	**63**	**60**
B5. Depreciation	35	44	45	45	44	42	41	40	38	36	34
B6. Combination of B1-B5	35	42	45	41	42	42	41	41	40	39	38
C. Tailored Tests											
C1. Combined contingent liabilities	35	45	48	50	51	51	50	49	48	47	45
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	126	140	149	162	163	163	170	166	160	154	148
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	126	112	94	79	57	36	18	(1)	(19)	(35)	(49)
B. Bound Tests											
B1. Real GDP growth	126	139	157	172	176	181	193	194	192	190	188
B2. Primary balance	126	160	195	208	208	208	215	210	203	196	189
B3. Exports	126	157	196	210	211	211	218	212	204	194	185
B4. Other flows 3/	126	186	240	256	257	255	264	256	244	231	219
B5. Depreciation	126	165	165	169	164	158	160	154	145	137	128
B6. Combination of B1-B5	126	151	159	147	147	148	155	152	148	143	139
C. Tailored Tests											
C1. Combined contingent liabilities	126	159	168	179	179	179	186	182	176	171	165
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	8	11	13	16	17	16	14	12	11	11	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	8	11	13	15	16	15	12	9	6	4	3
B. Bound Tests											
B1. Real GDP growth	8	11	14	17	18	18	15	14	13	13	13
B2. Primary balance	8	11	14	19	20	19	16	14	14	15	15
B3. Exports	8	11	13	17	18	18	15	13	13	14	14
B4. Other flows 3/	8	11	14	18	19	19	16	14	16	17	17
B5. Depreciation	8	12	16	19	21	21	17	15	14	14	14
B6. Combination of B1-B5	8	11	13	15	17	16	14	12	11	11	11
C. Tailored Tests											
C1. Combined contingent liabilities	8	11	14	17	18	18	15	13	12	12	12
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.